Lab Sheet 4 — SQL Operators

1. Perform addition of two numbers using +.

```
1 • SELECT 12 + 8 AS Addition;
```

Result Grid

Addition
20

2. Perform subtraction using -.

```
2 • SELECT 12 - 8 AS Subtraction;
```

Result Grid

Subtraction
4

3. Perform multiplication using *.

```
3 • SELECT 12 * 8 AS Multiplication;
```

Result Grid

Multiplication
96

4. Perform division using /.

```
4 • SELECT 12 / 8 AS Division;
```

Result Grid

Division
1.5000

5. Calculate the total amount as Price × Quantity.

```
5 • SELECT Product_Name, Price, Quantity, Price * Quantity AS Total_Amount FROM Product;
```

Result Grid

Product_Name	Price	Quantity	Total_Amount
Wooden Toy	450.00	10	4500.00
Clay Diya	80.00	25	2000.00
Handmade Vase	350.00	5	1750.00
Jute Bag	250.00	12	3000.00

6. Demonstrate AND with two conditions.

```
6 • SELECT * FROM Student WHERE Age >= 18 AND Address = 'Jaipur';
```

Result Grid

Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo	Dept_ID	Course_ID
101	Anshika	Female	20	Jaipur	B.Tech CSE	85.50	9876543210	1	201
105	Shreya	Female	20	Jaipur	B.Tech CSE	85.50	9876543276	1	201
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

7. Demonstrate OR with two conditions.

```
7 • SELECT * FROM Student WHERE Address = 'Jaipur' OR Address = 'Delhi';
```

Result Grid

Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo	Dept_ID	Course_ID
101	Anshika	Female	20	Jaipur	B.Tech CSE	85.50	9876543210	1	201
103	Jayshree	Female	17	Jaipur	B.Tech AIML	85.50	9876543276	1	202
105	Shreya	Female	20	Jaipur	B.Tech CSE	85.50	9876543276	1	201
107	Rahul	Male	21	Delhi	B.Tech CSE	78.25	9876543211	1	201
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

8. Demonstrate NOT with a condition.

```
8  SELECT * FROM Student WHERE NOT Address = 'Jaipur';
```

Result Grid

Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo	Dept_ID	Course_ID
102	Radha	Female	19	Kashipur	B.Tech CSE	85.50	9876543276	1	201
104	Ruhani	Female	20	kashmir	B.Tech CSE	85.50	9876543276	NULL	201
106	Maya	Female	20	rajasthan	B.Tech CSE	85.50	9876543276	NULL	201
107	Rahul	Male	21	Delhi	B.Tech CSE	78.25	9876543211	1	201
108	Priya	Female	20	Mumbai	BCA	91.00	9876543212	3	204
109	Aman	Male	22	Pune	B.Tech IT	74.75	9876543213	2	203
110	Neha	Female	21	Kolkata	B.Tech CSE	88.50	9876543214	1	201
111	Rohit	Male	20	Chandigarh	BCA	69.25	9876543215	3	204
112	Simran	Female	22	Amritsar	B.Tech IT	82.00	9876543216	2	203

9. Find students belonging to a specific course AND city.

```
9  SELECT * FROM Student WHERE Course = 'B.Tech CSE' AND Address = 'Jaipur';
```

Result Grid

Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo	Dept_ID	Course_ID
101	Anshika	Female	20	Jaipur	B.Tech CSE	85.50	9876543210	1	201
105	Shreya	Female	20	Jaipur	B.Tech CSE	85.50	9876543276	1	201

10. Find employees belonging to either of two departments.

```
10  SELECT * FROM Employee WHERE Department = 'IT' OR Department = 'HR';
```

Result Grid

Emp_ID	Emp_Name	Department	Salary	Joining_Date	Manager_ID	Dept_ID
2	Diya	HR	72000.50	2023-01-15	1	6
3	Kabir	IT	54000.00	2021-09-10	1	2
6	Sneha	IT	58250.25	2023-07-18	3	2

11. Demonstrate the = operator.

```
11  SELECT * FROM Student WHERE Age = 20;
```

Result Grid

Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo	Dept_ID	Course_ID
101	Anshika	Female	20	Jaipur	B.Tech CSE	85.50	9876543210	1	201
104	Ruhani	Female	20	kashmir	B.Tech CSE	85.50	9876543276	NULL	201
105	Shreya	Female	20	Jaipur	B.Tech CSE	85.50	9876543276	1	201
106	Maya	Female	20	rajasthan	B.Tech CSE	85.50	9876543276	NULL	201
108	Priya	Female	20	Mumbai	BCA	91.00	9876543212	3	204
111	Rohit	Male	20	Chandigarh	BCA	69.25	9876543215	3	204

12. Demonstrate the > operator.

```
12  SELECT * FROM Employee WHERE Salary > 60000;
```

Result Grid

Emp_ID	Emp_Name	Department	Salary	Joining_Date	Manager_ID	Dept_ID
1	Aarav	CSE	65000.00	2022-06-01	NULL	1
2	Diya	HR	72000.50	2023-01-15	1	6
4	Meera	Mechanical	61000.00	2020-03-20	1	4
7	Vikram	CSE	82000.00	2019-11-25	NULL	1

13. Demonstrate the < operator.

```
SELECT * FROM Student WHERE Age < 20;
```

Result Grid

Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo	Dept_ID	Course_ID
102	Radha	Female	19	Kashipur	B.Tech CSE	85.50	9876543276	1	201
103	Jayshree	Female	17	Jaipur	B.Tech AIML	85.50	9876543276	1	202

14. Demonstrate >= and <=.

```
SELECT * FROM Employee WHERE Salary >= 40000 AND Salary <= 70000;
```

Result Grid

Emp_ID	Emp_Name	Department	Salary	Joining_Date	Manager_ID	Dept_ID
1	Aarav	CSE	65000.00	2022-06-01	NULL	1
3	Kabir	IT	54000.00	2021-09-10	1	2
4	Meera	Mechanical	61000.00	2020-03-20	1	4
5	Rohan	CSE	48500.75	2024-02-05	1	1
6	Sneha	IT	58250.25	2023-07-18	3	2

15. Demonstrate the <>/!= operator.

```
SELECT * FROM Student WHERE Address <> 'Jaipur';
```

Result Grid

Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo	Dept_ID	Course_ID
102	Radha	Female	19	Kashipur	B.Tech CSE	85.50	9876543276	1	201
104	Ruhani	Female	20	kashmir	B.Tech CSE	85.50	9876543276	NULL	201
106	Maya	Female	20	rajasthan	B.Tech CSE	85.50	9876543276	NULL	201
107	Rahul	Male	21	Delhi	B.Tech CSE	78.25	9876543211	1	201
108	Priya	Female	20	Mumbai	BCA	91.00	9876543212	3	204
109	Aman	Male	22	Pune	B.Tech IT	74.75	9876543213	2	203
110	Neha	Female	21	Kolkata	B.Tech CSE	88.50	9876543214	1	201
111	Rohit	Male	20	Chandigarh	BCA	69.25	9876543215	3	204
112	Simran	Female	22	Amritsar	B.Tech IT	82.00	9876543216	2	203

16. Demonstrate BETWEEN for salary range.

```
SELECT * FROM Employee WHERE Salary BETWEEN 50000 AND 70000;
```

Result Grid

Emp_ID	Emp_Name	Department	Salary	Joining_Date	Manager_ID	Dept_ID
1	Aarav	CSE	65000.00	2022-06-01	NULL	1
3	Kabir	IT	54000.00	2021-09-10	1	2
4	Meera	Mechanical	61000.00	2020-03-20	1	4
6	Sneha	IT	58250.25	2023-07-18	3	2

17. Demonstrate IN for selecting multiple cities.

```
SELECT * FROM Student WHERE Address IN ('Jaipur', 'Delhi', 'Pune');
```

Result Grid

Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo	Dept_ID	Course_ID
101	Anshika	Female	20	Jaipur	B.Tech CSE	85.50	9876543210	1	201
103	Jayshree	Female	17	Jaipur	B.Tech AIML	85.50	9876543276	1	202
105	Shreya	Female	20	Jaipur	B.Tech CSE	85.50	9876543276	1	201
107	Rahul	Male	21	Delhi	B.Tech CSE	78.25	9876543211	1	201
109	Aman	Male	22	Pune	B.Tech IT	74.75	9876543213	2	203

18. Demonstrate NOT IN.

```
SELECT * FROM Student WHERE Address NOT IN ('Jaipur', 'Delhi', 'Pune');
```

Result Grid

Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo	Dept_ID	Course_ID
102	Radha	Female	19	Kashipur	B.Tech CSE	85.50	9876543276	1	201
104	Ruhani	Female	20	kashmir	B.Tech CSE	85.50	9876543276	NULL	201
106	Maya	Female	20	rajasthan	B.Tech CSE	85.50	9876543276	NULL	201
108	Priya	Female	20	Mumbai	BCA	91.00	9876543212	3	204
110	Neha	Female	21	Kolkata	B.Tech CSE	88.50	9876543214	1	201
111	Rohit	Male	20	Chandigarh	BCA	69.25	9876543215	3	204
112	Simran	Female	22	Amritsar	B.Tech IT	82.00	9876543216	2	203

19. Demonstrate LIKE for names beginning with a particular letter.

```
SELECT * FROM Student WHERE Name LIKE 'R%';
```

Result Grid

Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo	Dept_ID	Course_ID
102	Radha	Female	19	Kashipur	B.Tech CSE	85.50	9876543276	1	201
104	Ruhani	Female	20	kashmir	B.Tech CSE	85.50	9876543276	NULL	201
107	Rahul	Male	21	Delhi	B.Tech CSE	78.25	9876543211	1	201
111	Rohit	Male	20	Chandigarh	BCA	69.25	9876543215	3	204

20. Demonstrate IS NULL and IS NOT NULL.

Step 1

```
SELECT * FROM Employee WHERE Department IS NULL;
```

Result Grid

Emp_ID	Emp_Name	Department	Salary	Joining_Date	Manager_ID	Dept_ID
8	Pooja	NULL	35000.00	2024-08-12	4	NULL

Step 2

```
SELECT * FROM Employee WHERE Department IS NOT NULL;
```

Result Grid

Emp_ID	Emp_Name	Department	Salary	Joining_Date	Manager_ID	Dept_ID
1	Aarav	CSE	65000.00	2022-06-01	NULL	1
2	Diya	HR	72000.50	2023-01-15	1	6
3	Kabir	IT	54000.00	2021-09-10	1	2
4	Meera	Mechanical	61000.00	2020-03-20	1	4
5	Rohan	CSE	48500.75	2024-02-05	1	1
6	Sneha	IT	58250.25	2023-07-18	3	2
7	Vikram	CSE	82000.00	2019-11-25	NULL	1

21. Demonstrate UNION using two compatible queries.

```
SELECT Address AS Place FROM Student
UNION
SELECT Location FROM Department;
```

Result Grid

Place
Jaipur
Kashipur
kashmir
rajasthan
Delhi
Mumbai
Pune
Kolkata
Chandigarh
Amritsar
Haridwar
Meerut
Roorkee

22. Demonstrate UNION ALL.

```
25  SELECT Address AS Place FROM Student
26  UNION ALL
27  SELECT Location FROM Department;
```

Result Grid

Place
Jaipur
Kashipur
Jaipur
kashmir
Jaipur
rajasthan
Delhi
Mumbai
Pune
Kolkata
Chandigarh
Amritsar
Jaipur
Pune
Delhi

23. Demonstrate INTERSECT where supported by the DBMS.

```
28  SELECT DISTINCT Address FROM Student
29  WHERE Address IN (SELECT Location FROM Department);
```

Result Grid

Address
Jaipur
Delhi
Pune

24. Demonstrate EXCEPT/MINUS where supported.

```
30  SELECT DISTINCT Address FROM Student
31  WHERE Address NOT IN (SELECT Location FROM Department);
```

Result Grid

Address
Kashipur
kashmir
rajasthan
Mumbai
Kolkata
Chandigarh
Amritsar

25. Perform a suitable problem using multiple operators together, such as Arithmetic + Logical operators.

```
32  SELECT Name, Age, Percentage, Percentage + 5 AS Percentage_With_Grace FROM Student
33  WHERE Percentage + 5 > 85 AND (Course = 'B.Tech CSE' OR Course = 'BCA') AND NOT Address = 'Mumbai';
```

Result Grid

Name	Age	Percentage	Percentage_With_Grace
Anshika	20	85.50	90.50
Radha	19	85.50	90.50
Ruhani	20	85.50	90.50
Shreya	20	85.50	90.50
Maya	20	85.50	90.50
Neha	21	88.50	93.50